Applied Physics - 1

Motion in One Dimension - Lecture 03 (Module -01)

Motion in One Dimension

Motion is the change in the position of an object with respect to a reference point as time changes. A body is said to be in motion when its position changes with time relative to a chosen reference frame.
The study of motion without considering the causes producing the motion is called **kinematics**. Kinematics describes the position, displacement, velocity, acceleration, and other quantities associated with the motion of an object.
When an object moves along a straight-line path, its motion can be described using a single spatial coordinate, usually denoted by x. Such motion is called **one-dimensional motion** or **rectilinear motion**.
Examples of one-dimensional motion include the motion of a car along a straight road, the vertical motion of a lift, the motion of a train along a straight track, and the motion of an object falling vertically.
The position of an object in one-dimensional motion can be represented as a function of time:

$$\boxed{x = x(t)}$$

where x represents the position of the object and t represents time.

Reference Point and Position

To describe the motion of an object, a reference point or origin must first be selected. The position of an object is then specified relative to this reference point.
Consider a straight-line coordinate system with the origin at

$$x = 0.$$

Positions in one direction are assigned positive values and positions in the opposite direction are assigned negative values.
Thus,

$$x > 0$$

represents a position in the positive direction, while

$$x < 0$$

represents a position in the negative direction.
The position of an object therefore specifies its location relative to the chosen origin.
The position may change with time and can be represented by

$$\boxed{x = x(t)}$$

The change in position is used to define displacement.

Distance

Distance is the total length of the actual path travelled by an object during its motion.
Distance is a **scalar quantity** because it has magnitude only and does not have a direction.

The SI unit of distance is metre:

$$\boxed{\text{m}}$$

If an object travels through successive path lengths $d_1, d_2, d_3, \ldots, d_n$, the total distance travelled is

$$\boxed{d = d_1 + d_2 + d_3 + \cdots + d_n}$$

or

$$\boxed{d = \sum_{i=1}^{n} d_i}$$

Since distance represents the total length of the path travelled,

$$\boxed{d \geq 0}$$

Distance depends on the actual path followed by the object. Therefore, two objects moving between the same initial and final positions may have different distances if they follow different paths.

Distance in One-Dimensional Motion

For motion along a straight line, if the object moves continuously in one direction, the distance travelled is equal to the magnitude of the change in position.
If the object moves from x_i to x_f without changing direction,

$$\boxed{d = |x_f - x_i|}$$

If the object changes its direction during motion, the total distance must be calculated by adding the lengths of all individual parts of the path.
For example, suppose a particle moves 8 m in the positive direction and then moves 3 m in the negative direction.
The total distance is

$$d = 8 + 3.$$

Therefore,

$$\boxed{d = 11\,\text{m}}$$

Displacement

Displacement is the change in the position of an object from its initial position to its final position.
Unlike distance, displacement is a **vector quantity**. Therefore, it has both magnitude and direction.
The displacement vector is defined as

$$\boxed{\Delta\vec{r} = \vec{r}_f - \vec{r}_i}$$

where $\vec{r}_i$ is the initial position vector and $\vec{r}_f$ is the final position vector.
For one-dimensional motion along the x-axis,

$$\boxed{\Delta x = x_f - x_i}$$

The SI unit of displacement is

$$\boxed{\text{m}}$$

The sign of displacement indicates the direction of the change in position relative to the selected coordinate system.

Positive Displacement

If the final position is greater than the initial position,

$$x_f > x_i,$$

then

$$\Delta x > 0.$$

The displacement is therefore in the positive direction.

Negative Displacement

If

$$x_f < x_i,$$

then

$$\Delta x < 0.$$

The displacement is therefore in the negative direction.

Zero Displacement

If the final position is equal to the initial position,

$$x_f = x_i,$$

then

$$\boxed{\Delta x = 0}$$

An object can have zero displacement even though it has travelled a non-zero distance. This occurs when the object returns to its initial position.

Relationship Between Distance and Displacement

The magnitude of displacement can never be greater than the total distance travelled.
Therefore,

$$\boxed{d \geq |\Delta x|}$$

Equality occurs when the object moves along a straight line without changing its direction.
For a complete round trip,

$$\boxed{\Delta x = 0}$$

while

$$\boxed{d > 0}$$

Thus, distance describes the total path travelled, whereas displacement describes the net change in position.

Distance and Displacement Comparison

Distance	Displacement
Scalar quantity	Vector quantity
Magnitude only	Magnitude and direction
Depends on actual path	Depends on initial and final positions
Always non-negative	Can be positive, negative, or zero
$d \geq 0$	$\Delta x = x_f - x_i$
$d \geq \lvert\Delta x\rvert$	$\lvert\Delta x\rvert \leq d$

Speed

Speed describes how rapidly an object covers distance with time.
It is defined as the rate of change of distance with respect to time.
The average speed of an object is

$$\boxed{v_{\text{avg}} = \frac{\text{Total Distance}}{\text{Total Time}}}$$

Therefore,

$$\boxed{v_{\text{avg}} = \frac{d}{t}}$$

where d is the total distance travelled and t is the total time taken.
The SI unit of speed is

$$\boxed{\text{m/s}}$$

Speed is a scalar quantity and therefore does not specify the direction of motion.

Instantaneous Speed

Instantaneous speed represents the speed of an object at a particular instant of time.
If s represents the distance travelled, instantaneous speed is

$$\boxed{v = \frac{ds}{dt}}$$

Since speed is the magnitude of velocity,

$$\boxed{v = |\vec{v}|}$$

Instantaneous speed is therefore always non-negative:

$$\boxed{v \geq 0}$$

Velocity

Velocity describes the rate at which the displacement of an object changes with time.
Since displacement is a vector quantity, velocity is also a **vector quantity**.
The average velocity is defined as

$$\boxed{\vec{v}_{\text{avg}} = \frac{\Delta \vec{r}}{\Delta t}}$$

For one-dimensional motion,

$$\boxed{v_{\text{avg}} = \frac{\Delta x}{\Delta t}}$$

where

$$\Delta x = x_f - x_i$$

and

$$\Delta t = t_f - t_i.$$

The SI unit of velocity is

$$\boxed{\text{m/s}}$$

The sign of velocity indicates the direction of motion relative to the chosen coordinate system.

Instantaneous Velocity

Instantaneous velocity is the velocity of an object at a particular instant of time.
It is obtained by considering the limiting value of average velocity as the time interval approaches zero.
Therefore,

$$\boxed{\vec{v} = \lim_{\Delta t \to 0} \frac{\Delta \vec{r}}{\Delta t}}$$

Using differential notation,

$$\boxed{\vec{v} = \frac{d\vec{r}}{dt}}$$

For one-dimensional motion,

$$\boxed{v = \frac{dx}{dt}}$$

Thus, velocity is the first derivative of position with respect to time.

Relationship Between Speed and Velocity

Instantaneous Speed is the magnitude of Instantaneous velocity.
Therefore,

$$\boxed{\text{Speed} = |\text{Velocity}|}$$

or

$$\boxed{v_{\text{speed}} = |\vec{v}|}$$

For average quantities,

$$\boxed{\text{Average Speed} \geq |\text{Average Velocity}|}$$

Equality occurs when the object moves along a straight path without changing its direction.

Acceleration

Acceleration describes the rate at which velocity changes with time.
An object is accelerated whenever its velocity changes in magnitude, direction, or both.
The average acceleration is defined as

$$\boxed{\vec{a}_{\text{avg}} = \frac{\Delta \vec{v}}{\Delta t}}$$

For one-dimensional motion,

$$\boxed{a_{\text{avg}} = \frac{v_f - v_i}{t_f - t_i}}$$

The SI unit of acceleration is

$$\boxed{\text{m/s}^2}$$

Instantaneous Acceleration

Instantaneous acceleration is the acceleration of an object at a particular instant.
It is defined as

$$\boxed{\vec{a} = \frac{d\vec{v}}{dt}}$$

Since

$$\vec{v} = \frac{d\vec{r}}{dt},$$

we obtain

$$\boxed{\vec{a} = \frac{d^2\vec{r}}{dt^2}}$$

For one-dimensional motion,

$$\boxed{a = \frac{dv}{dt}}$$

and

$$\boxed{a = \frac{d^2x}{dt^2}}$$

Thus, acceleration is the second derivative of position with respect to time.

Positive Acceleration

If acceleration acts in the positive direction,

$$\boxed{a > 0}$$

with respect to the chosen coordinate system.

Negative Acceleration

If acceleration acts in the negative direction,

$$\boxed{a < 0}$$

A negative acceleration does not necessarily mean that the object is slowing down. The effect of acceleration on speed depends on the relative directions of velocity and acceleration.
If velocity and acceleration have the same direction, the speed increases.
If velocity and acceleration have opposite directions, the speed decreases.

Jerk

Jerk describes the rate at which acceleration changes with time.
It is defined as the time derivative of acceleration.
The instantaneous jerk is

$$\boxed{j = \frac{da}{dt}}$$

Since

$$a = \frac{dv}{dt},$$

jerk can also be written as

$$\boxed{j = \frac{d^2v}{dt^2}}$$

Since

$$v = \frac{dx}{dt},$$

we obtain

$$\boxed{j = \frac{d^3x}{dt^3}}$$

The SI unit of jerk is

$$\boxed{\mathrm{m/s^3}}$$

Jerk describes how rapidly the acceleration experienced by an object changes. Smooth changes in acceleration correspond to smaller jerk, while sudden changes in acceleration correspond to larger jerk.
Jerk is therefore considered in systems where smooth motion is required, including elevators, vehicles, robotic mechanisms, trains, and other mechanical systems.

Motion Variables and Their Rates of Change

The successive derivatives of position describe different quantities associated with motion.

$$\boxed{x \xrightarrow{\frac{d}{dt}} v \xrightarrow{\frac{d}{dt}} a \xrightarrow{\frac{d}{dt}} j}$$

Therefore,

$$\boxed{v = \frac{dx}{dt}}$$

$$\boxed{a = \frac{dv}{dt} = \frac{d^2x}{dt^2}}$$

and

$$\boxed{j = \frac{da}{dt} = \frac{d^2v}{dt^2} = \frac{d^3x}{dt^3}}$$

Uniform and Non-Uniform Motion

Uniform Motion

An object is said to be in uniform motion when it undergoes equal displacements in equal intervals of time.
For uniform motion, velocity remains constant:

$$\boxed{v = \text{constant}}$$

Therefore,

$$\boxed{a = 0}$$

If the initial position is x_0, the position after time t is

$$\boxed{x = x_0 + vt}$$

The displacement is

$$\boxed{\Delta x = vt}$$

Uniform motion therefore corresponds to constant velocity and zero acceleration.

Non-Uniform Motion

An object is said to be in non-uniform motion when its velocity changes with time.
Therefore,

$$\boxed{v \neq \text{constant}}$$

and generally,

$$\boxed{a \neq 0}$$

The velocity may change because the object changes its speed, direction, or both.

Graphical Understanding of Motion

Graphs provide a convenient method for representing the variation of motion quantities with time.
The three fundamental graphs used in one-dimensional kinematics are:

1. Position–time graph
2. Velocity–time graph
3. Acceleration–time graph

These graphs provide information about position, velocity, acceleration, displacement, and change in velocity.

Position–Time Graph

Position–Time Graph

A position–time graph represents the position of an object as a function of time. Position x is plotted along the vertical axis, while time t is plotted along the horizontal axis.
The slope of a position–time graph represents the velocity of the object. For a finite time interval, the average slope is

$$\boxed{\text{Slope} = \frac{\Delta x}{\Delta t}}$$

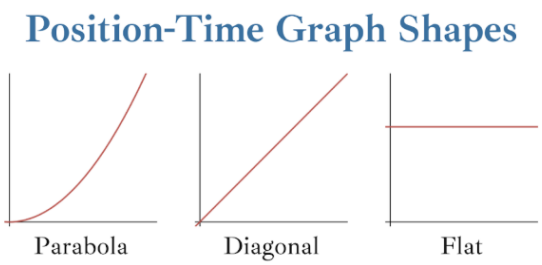
Since

$$v_{\text{avg}} = \frac{\Delta x}{\Delta t},$$

the average slope of a position–time graph represents the average velocity.
For instantaneous velocity, the slope of the tangent to the position–time curve at a particular instant is

$$\boxed{v = \frac{dx}{dt}}$$

Thus, the shape and slope of the position–time graph indicate how the velocity changes with time.



Parabolic Position–Time Graph

In the **Parabola** shown in the Figure, the position does not increase at a constant rate. As time increases, the curve becomes progressively steeper.
Since the slope of the position–time graph represents velocity,

$$v = \frac{dx}{dt},$$

an increasing slope means that the velocity is increasing with time.
Therefore,

$$\boxed{v = \text{variable}}$$

and the object is accelerating.
For constant acceleration, the position is given by

$$x = x_0 + ut + \frac{1}{2}at^2.$$

Differentiating with respect to time,

$$v = \frac{dx}{dt} = u + at.$$

Thus, the velocity changes continuously as time increases.
For the parabolic curve shown, the slope is small at earlier times and becomes larger at later times. Hence,

$$\boxed{a > 0}$$

for the motion represented by the curve.

Straight-Line Position–Time Graph

In the **Diagonal** graph shown in the Figure, the position increases linearly with time.
The graph has the same slope at every instant. Therefore,

$$\frac{\Delta x}{\Delta t} = \text{constant}.$$

Hence,

$$\boxed{v = \text{constant}}$$

The object covers equal displacements in equal intervals of time.
Since velocity does not change with time,

$$a = \frac{dv}{dt} = 0.$$

Therefore,

$$\boxed{a = 0}$$

The object undergoes uniform motion.
The equation of position for such motion is

$$\boxed{x = x_0 + vt}$$

where x_0 is the initial position and v is the constant velocity.

Horizontal Position–Time Graph

In the **Flat** graph shown in the Figure, the position remains unchanged as time increases.
The graph is horizontal, so its slope is zero:

$$\frac{\Delta x}{\Delta t} = 0.$$

Therefore,

$$\boxed{v = 0}$$

The position of the object remains constant throughout the corresponding time interval.
Thus, the object is at rest.
For example, if the graph remains horizontal between times t_1 and t_2,

$$x(t) = x_0, \qquad t_1 \le t \le t_2.$$

Hence,

$$\boxed{v = 0}$$

throughout this interval.

Position–Time Graph Interpretation with Time

The position–time graph must always be interpreted from left to right, because the horizontal axis represents increasing time.

- **Parabola:** The slope increases with time, so the velocity changes with time and the object is accelerated.
- **Diagonal:** The slope remains constant with time, so the velocity remains constant and the object undergoes uniform motion.
- **Flat:** The slope is zero at all times, so the velocity is zero and the object remains at rest.

Thus, the position–time graph provides a direct graphical method for determining the velocity and nature of motion of an object:

$$\boxed{\text{Slope of } x-t \text{ graph} = \text{Velocity}}$$

and

$$\boxed{\text{Change in slope with time} = \text{Change in velocity}}$$

Therefore, by observing how the slope of the position–time graph changes as time progresses, the motion of the object can be identified.

Velocity–Time Graph

A velocity–time graph represents the variation of velocity of an object with time. Velocity v is plotted along the vertical axis, while time t is plotted along the horizontal axis.
The slope of a velocity–time graph represents the acceleration of the object. For a finite time interval, the average slope is

$$\boxed{\text{Slope} = \frac{\Delta v}{\Delta t}}$$

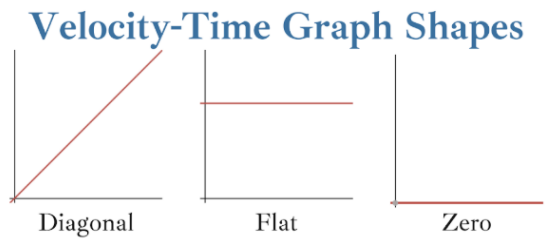
Since

$$a_{\text{avg}} = \frac{\Delta v}{\Delta t},$$

the average slope of a velocity–time graph represents the average acceleration.
For instantaneous acceleration, the slope of the tangent to the velocity–time graph at a particular instant is

$$\boxed{a = \frac{dv}{dt}}$$

Thus, the shape and slope of a velocity–time graph indicate how the acceleration changes with time.



Diagonal Velocity–Time Graph

In the **Diagonal** graph shown in Fig, the velocity changes continuously as time increases.
The graph is a straight line, so its slope remains constant throughout the motion. Since the slope of a velocity–time graph represents acceleration,

$$\frac{\Delta v}{\Delta t} = \text{constant}.$$

Therefore,

$$\boxed{a = \text{constant}}$$

The object undergoes uniformly accelerated motion.
In the graph shown, the velocity increases as time increases. Therefore, the slope is positive and

$$\boxed{a > 0}$$

For constant acceleration, the velocity is given by

$$\boxed{v = u + at}$$

where u is the initial velocity and a is the constant acceleration.
Thus, equal intervals of time produce equal changes in velocity:

$$\Delta v_1 = \Delta v_2 = \Delta v_3$$

when

$$\Delta t_1 = \Delta t_2 = \Delta t_3.$$

If the diagonal line slopes downward instead of upward, the velocity decreases with time and the acceleration is negative:

$$\boxed{a < 0}$$

Horizontal Velocity–Time Graph

In the **Flat** graph shown in Fig, the velocity remains unchanged as time increases.
Since the graph is horizontal, its slope is zero:

$$\frac{\Delta v}{\Delta t} = 0.$$

Therefore,

$$\boxed{a = 0}$$

The velocity remains constant throughout the corresponding time interval.
For example, if the graph remains horizontal from t_1 to t_2,

$$v(t) = v_0, \qquad t_1 \le t \le t_2.$$

Hence,

$$\boxed{v = \text{constant}}$$

and

$$\boxed{a = 0.}$$

The object is therefore undergoing uniform motion.
The displacement during this time interval is equal to the area under the velocity–time graph. For constant velocity,

$$\boxed{\Delta x = v(t_2 - t_1)}$$

Thus, the rectangular area between the velocity line and the time axis represents the displacement.

Zero Velocity–Time Graph

In the **Zero** graph shown in Fig, the velocity remains equal to zero throughout the time interval.
Therefore,

$$\boxed{v = 0}$$

and the object remains at rest.
Since the velocity does not change with time,

$$\frac{dv}{dt} = 0,$$

so

$$\boxed{a = 0.}$$

If the velocity remains zero from t_1 to t_2, then

$$v(t) = 0, \qquad t_1 \le t \le t_2.$$

The displacement during this interval is also zero because there is no velocity:

$$\boxed{\Delta x = 0}$$

Area Under a Velocity–Time Graph

The area under a velocity–time graph represents the displacement of the object during the corresponding time interval.
For a small time interval dt, the small displacement is

$$d(x) = v\,dt.$$

Integrating between the initial time t_i and final time t_f,

$$\boxed{\Delta x = \int_{t_i}^{t_f} v(t)\,dt}$$

Therefore, the algebraic area under the velocity–time graph gives the displacement.
For constant velocity, the graph forms a rectangle. Hence,

$$\Delta x = \text{base} \times \text{height}$$

and therefore,

$$\boxed{\Delta x = vt}$$

For constant acceleration, the velocity–time graph is a straight diagonal line and the area under the graph forms a trapezium. Thus,

$$\Delta x = \frac{1}{2}(u + v)t.$$

Therefore,

$$\boxed{\Delta x = \frac{u+v}{2}t}$$

If the velocity–time graph consists of several simple geometric regions, the total displacement is obtained by adding the algebraic areas of all the regions.

Velocity–Time Graph Interpretation with Time

The velocity–time graph is interpreted from left to right, with time increasing along the horizontal axis.
For the three shapes shown in Fig. **??**:

- **Diagonal:** Velocity changes with time. Since the slope is constant, the object has constant acceleration. In the graph shown, velocity increases with time and therefore $a > 0$.
- **Flat:** Velocity remains constant as time increases. The slope is zero, so $a = 0$. The object undergoes uniform motion.
- **Zero:** Velocity remains zero with time. The object is at rest and both velocity and acceleration are zero during the corresponding interval.

Thus, a velocity–time graph provides two important pieces of information:

$$\boxed{\text{Slope of } v - t \text{ graph} = \text{Acceleration}}$$

and

$$\boxed{\text{Area under } v - t \text{ graph} = \text{Displacement}}$$

Hence, by observing how the velocity changes with time and by examining the slope and area of the graph, the acceleration and displacement of the object can be determined.

Acceleration–Time Graph

An acceleration–time graph represents the variation of acceleration of an object with time. Acceleration a is plotted along the vertical axis, while time t is plotted along the horizontal axis.
The area under an acceleration–time graph represents the change in velocity of the object.
For a small time interval dt,

$$dv = a\,dt.$$

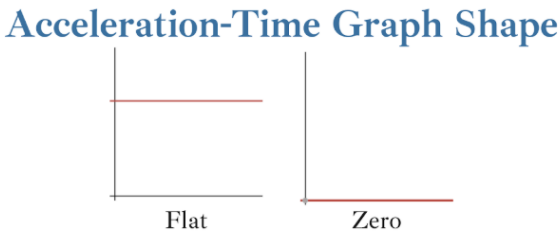
Integrating between the initial time t_i and final time t_f,

$$\boxed{\Delta v = \int_{t_i}^{t_f} a(t)\,dt}$$

Therefore,

$$\boxed{v_f - v_i = \int_{t_i}^{t_f} a(t)\,dt}$$

Thus, the algebraic area under an acceleration–time graph gives the change in velocity during the corresponding time interval.



Flat Acceleration–Time Graph

In the **Flat** graph shown in Fig, the acceleration remains constant as time increases.
Since the graph is horizontal, its value does not change with time:

$$\boxed{a = \text{constant}}$$

For example, if the acceleration remains constant from time t_1 to t_2,

$$a(t) = a_0, \qquad t_1 \le t \le t_2.$$

The change in velocity during this interval is obtained from the area under the graph:

$$\Delta v = a_0(t_2 - t_1).$$

Therefore,

$$\boxed{v_2 - v_1 = a_0(t_2 - t_1)}$$

If the initial time is taken as 0 and the final time as t, this becomes

$$\boxed{v_f - v_i = at}$$

and hence,

$$\boxed{v_f = v_i + at}$$

Thus, a horizontal acceleration–time graph represents uniformly accelerated motion.

Positive Constant Acceleration

If the horizontal acceleration line lies above the time axis, the acceleration is positive:

$$\boxed{a > 0}$$

As time increases, the velocity increases at a constant rate. For equal time intervals,

$$\Delta t_1 = \Delta t_2 = \Delta t_3,$$

the corresponding changes in velocity are equal:

$$\Delta v_1 = \Delta v_2 = \Delta v_3.$$

Therefore, positive constant acceleration produces a uniform increase in velocity with time.

Negative Constant Acceleration

If a horizontal acceleration line lies below the time axis, the acceleration is negative:

$$\boxed{a < 0}$$

In this case, the velocity changes at a constant negative rate. For a time interval from t_1 to t_2,

$$v_2 - v_1 = a(t_2 - t_1).$$

Since $a < 0$,

$$v_2 < v_1$$

provided the object continues to move in the positive direction during the interval.

Thus, negative acceleration corresponds to a decreasing velocity when the velocity is positive. However, negative acceleration should not automatically be interpreted as deceleration, because acceleration and velocity may have different directions.

Zero Acceleration–Time Graph

In the **Zero** graph shown in Fig, the acceleration remains equal to zero as time increases.
Therefore,

$$\boxed{a = 0}$$

Since

$$a = \frac{dv}{dt},$$

we have

$$\frac{dv}{dt} = 0.$$

Hence, the velocity remains constant:

$$\boxed{v = \text{constant}}$$

Thus, zero acceleration means that the velocity does not change with time. It does not necessarily mean that the object is at rest.

For example, an object moving with a constant velocity

$$v = 5\,\text{m/s}$$

has

$$a = 0.$$

The object continues to move uniformly.

If the acceleration is zero from t_1 to t_2, then

$$\Delta v = \int_{t_1}^{t_2} 0\,dt = 0.$$

Therefore,

$$\boxed{v_2 = v_1}$$

and the velocity at the beginning and end of the interval is the same.

Area Under an Acceleration–Time Graph

The area under an acceleration–time graph represents the change in velocity.
For constant acceleration, the graph forms a rectangle. Therefore,

$$\Delta v = \text{base} \times \text{height}$$

$$\Delta v = t \times a.$$

Hence,

$$\boxed{\Delta v = at}$$

or

$$\boxed{v_f - v_i = at}$$

which gives the first equation of motion:

$$\boxed{v_f = v_i + at}$$

If the acceleration is positive, the area above the time axis gives a positive change in velocity.
If the acceleration is negative, the area below the time axis gives a negative change in velocity.
If the acceleration is zero, the area is zero and there is no change in velocity.

Acceleration–Time Graph Interpretation with Time

The acceleration–time graph is interpreted from left to right, with time increasing along the horizontal axis.
For the two shapes shown in Fig:

- **Flat:** Acceleration remains constant as time increases. The velocity therefore changes uniformly with time.
- **Zero:** Acceleration remains zero as time increases. The velocity therefore remains constant.

Thus, an acceleration–time graph provides direct information about the change in velocity:

$$\boxed{\text{Area under } a-t \text{ graph} = \text{Change in velocity}}$$

and

$$\boxed{\text{Slope of } a-t \text{ graph} = \text{Jerk}}$$

where

$$\boxed{j = \frac{da}{dt}}$$

Therefore, by observing the acceleration at successive instants of time and calculating the area under the acceleration–time graph, the change in velocity of the object can be determined.

Relationships Between Motion Graphs

The fundamental slope relationships are

$$\boxed{\text{Slope of } x-t \text{ graph} = v}$$

$$\boxed{\text{Slope of } v-t \text{ graph} = a}$$

$$\boxed{\text{Slope of } a-t \text{ graph} = j}$$

The corresponding area relationships are

$$\boxed{\text{Area under } v-t \text{ graph} = \Delta x}$$

$$\boxed{\text{Area under } a-t \text{ graph} = \Delta v}$$

$$\boxed{\text{Area under } j-t \text{ graph} = \Delta a}$$

The complete derivative relationship is

$$\boxed{x \xrightarrow{\frac{d}{dt}} v \xrightarrow{\frac{d}{dt}} a \xrightarrow{\frac{d}{dt}} j}$$

The corresponding integral relationship is

$$\boxed{j \xrightarrow{\int dt} a \xrightarrow{\int dt} v \xrightarrow{\int dt} x}$$

Motion in a Straight Line

Motion in which an object moves along a straight path is called **straight-line motion**, **rectilinear motion**, or **one-dimensional motion**.
The position of the object can be represented using one coordinate, usually x.
The displacement is

$$\boxed{\Delta x = x_f - x_i}$$

The velocity is

$$\boxed{v = \frac{dx}{dt}}$$

The acceleration is

$$\boxed{a = \frac{dv}{dt} = \frac{d^2x}{dt^2}}$$

and the jerk is

$$\boxed{j = \frac{da}{dt} = \frac{d^3x}{dt^3}}$$

Motion with Constant Acceleration

A body is said to have constant acceleration when its velocity changes by equal amounts in equal intervals of time.
Mathematically,

$$\boxed{a = \text{constant}}$$

For constant acceleration,

$$a = \frac{v-u}{t}$$

where

$$u = \text{initial velocity},$$

$$v = \text{final velocity},$$

$$a = \text{constant acceleration},$$

and

$$t = \text{time interval}.$$

Constant-acceleration motion is described by the standard equations of motion.

Equations of Motion

For one-dimensional motion with constant acceleration, let

$$u = \text{initial velocity},$$

$$v = \text{final velocity},$$

$$a = \text{constant acceleration},$$

$$t = \text{time},$$

and

$$s = \text{displacement}.$$

First Equation of Motion

Acceleration is defined as the rate of change of velocity:

$$a = \frac{\text{change in velocity}}{\text{time}}.$$

Therefore,

$$a = \frac{v - u}{t}.$$

Multiplying both sides by t,

$$at = v - u.$$

Rearranging,

$$\boxed{v = u + at}$$

This is the first equation of motion.
The equation relates initial velocity, final velocity, acceleration, and time.
The rearranged forms are

$$\boxed{u = v - at}$$

$$\boxed{a = \frac{v - u}{t}}$$

and

$$\boxed{t = \frac{v - u}{a}}$$

Second Equation of Motion

For constant acceleration, the average velocity is the arithmetic mean of the initial and final velocities.
Therefore,

$$\boxed{v_{\text{avg}} = \frac{u + v}{2}}$$

Displacement is given by

$$s = v_{\text{avg}} t.$$

Hence,

$$s = \frac{u + v}{2} t.$$

Using the first equation of motion,

$$v = u + at.$$

Substituting,

$$s = \frac{u + (u + at)}{2} t.$$

Therefore,

$$s = \frac{2u + at}{2} t.$$

Simplifying,

$$\boxed{s = ut + \frac{1}{2} a t^2}$$

This is the second equation of motion.

Third Equation of Motion

The first equation of motion is

$$v = u + at.$$

Rearranging,

$$t = \frac{v - u}{a}.$$

Using

$$s = \frac{u + v}{2} t,$$

we obtain

$$s = \frac{u + v}{2} \left(\frac{v - u}{a} \right).$$

Therefore,

$$s = \frac{(u + v)(v - u)}{2a}.$$

Using the identity

$$(a + b)(a - b) = a^2 - b^2,$$

we obtain

$$s = \frac{v^2 - u^2}{2a}.$$

Multiplying by $2a$,

$$2as = v^2 - u^2.$$

Hence,

$$\boxed{v^2 = u^2 + 2as}$$

This is the third equation of motion.

Special Cases of Equations of Motion

Motion Starting From Rest

When an object starts from rest,

$$\boxed{u = 0}$$

The first equation of motion becomes

$$v = at.$$

Therefore,

$$\boxed{v = at}$$

The second equation becomes

$$s = \frac{1}{2}at^2.$$

Therefore,

$$\boxed{s = \frac{1}{2}at^2}$$

The third equation becomes

$$v^2 = 2as.$$

Therefore,

$$\boxed{v^2 = 2as}$$

Motion With Zero Acceleration

If

$$a = 0,$$

the first equation becomes

$$v = u.$$

Therefore,

$$\boxed{v = u = \text{constant}}$$

The second equation becomes

$$s = ut.$$

Since $u = v$,

$$\boxed{s = vt}$$

This represents uniform motion along a straight line.

Retardation

When acceleration acts opposite to the direction of velocity, the magnitude of velocity decreases. Such acceleration is commonly called retardation or deceleration.
If the positive direction is chosen along the initial direction of motion, retardation is represented by

$$\boxed{a < 0}$$

The equations of motion remain unchanged:

$$\boxed{v = u + at}$$

$$\boxed{s = ut + \frac{1}{2}at^2}$$

$$\boxed{v^2 = u^2 + 2as}$$

The negative value of a automatically accounts for the direction of the acceleration.

Worksheet Questions

1. A car travels 150 m in 10 s along a straight road. Determine its average speed in m/s and km/h.

2. A student walks 120 m east and then 50 m west along the same straight path. Determine the total distance travelled and the displacement of the student.

3. A car starts from rest and moves with a constant acceleration of $3\,\text{m/s}^2$ for 8 s. Determine its final velocity and the displacement covered during this interval.

4. A particle has an initial velocity of 12 m/s and accelerates uniformly at $2.5\,\text{m/s}^2$ for 6 s. Determine its final velocity and displacement.

5. The velocity of a particle changes uniformly from 8 m/s to 32 m/s in 6 s. Determine the acceleration and displacement of the particle during this interval.

6. The position of a particle moving along a straight line is given by

$$x(t) = 3t^3 - 6t^2 + 4t + 2$$

where x is in metres and t is in seconds. Determine the velocity and acceleration at $t = 2\,\text{s}$.

7. A particle moves with an initial velocity of $20\,\text{m/s}$ and a constant acceleration of $-4\,\text{m/s}^2$. Determine the time at which it comes momentarily to rest and the displacement covered before coming to rest.

8. A velocity–time graph consists of a straight line increasing from $0\,\text{m/s}$ at $t = 0$ to $24\,\text{m/s}$ at $t = 6\,\text{s}$, followed by a constant velocity of $24\,\text{m/s}$ for the next $4\,\text{s}$. Determine the total displacement and the average velocity over the entire $10\,\text{s}$ interval.

9. The velocity of a particle is given by

$$v(t) = 4t^2 - 6t + 5$$

where v is in m/s and t is in seconds. Determine the acceleration at $t = 3\,\text{s}$ and the displacement of the particle from $t = 0$ to $t = 3\,\text{s}$.

10. The acceleration of a particle varies uniformly from $2\,\mathrm{m/s^2}$ at $t = 0$ to $8\,\mathrm{m/s^2}$ at $t = 6\,\mathrm{s}$. If its initial velocity is $5\,\mathrm{m/s}$, determine the change in velocity and final velocity using the area under the acceleration–time graph.

11. A robot's end-effector moves along a straight axis according to

$$x(t) = 2t^3 - 9t^2 + 12t$$

where x is in metres and t is in seconds. Determine the velocity, acceleration, and jerk as functions of time. Also determine the instants at which the end-effector is momentarily at rest and the displacement between those instants.

12. A particle moves along a straight line with acceleration

$$a(t) = 4t - 2$$

where a is in $\mathrm{m/s^2}$ and t is in seconds. At $t = 0$, its velocity is $3\,\mathrm{m/s}$ and its position is $2\,\mathrm{m}$. Determine the velocity function, position function, velocity at $t = 4\,\mathrm{s}$, displacement during the first $4\,\mathrm{s}$, and jerk.